Amirreza Mohammadnezhad Shebly

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Research Interests

- Computer Vision
- Generative Modeling
- Autonomous Driving
- Uncertainty Estimation
- Trajectory Prediction
- Video Understanding

Education

Bilkent University *Sep 2025 – Present*
M.Sc. in Computer Engineering Ankara, Türkiye
GPA: 3.53/4

University of Tabriz *Sep 2021 – Sep 2025*
B.Sc. in Computer Engineering Tabriz, Iran

GPA: 17.54/20 | Official Transcript-reported GPA Rank: 1/44

Publications

PAMR: Precedence-Aware Mode Repair for Multimodal Trajectory Forecasting
Amirreza M. Shebly, Ehsan Garaaghaji, Doruk Öner
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2027**).*

Direction-Aware Kinematics-Conditioned Graph Hypernetwork for Human Trajectory Prediction
Amirreza M. Shebly, Mohammad Sarhangzadeh, Doruk Öner
*Asian Conference on Computer Vision (**ACCV 2026**) — **Oral Presentation**.*

Stochastic Nonlinearities Improve Uncertainty Estimation
Mohammad Sarhangzadeh, **Amirreza M. Shebly**, Doruk Öner
*British Machine Vision Conference (**BMVC 2026**).*

Manuscripts and Ongoing Research

Trajectory Acquisition Variability in Pedestrian Crossing-Intention Prediction
Research in progress.

TopoRefiner: Topology-Aware Refinement of Curvilinear Structure Segmentation
Ehsan Garaaghaji, **Amirreza M. Shebly**, Doruk Öner
Manuscript in preparation.

Research Experience

NeuraVision Lab, Bilkent University *Sep 2025 – Present*
Researcher Advisor: Doruk Öner

2025–2026 **Direction-Aware Kinematics-Conditioned Graph Hypernetwork for Human Trajectory Prediction**

- Developed a lightweight trajectory forecasting framework based on a sparse dynamic interaction graph, where relative position, velocity, directional alignment, and time-to-collision cues determine physically meaningful agent interactions.
- Designed a kinematics-conditioned graph hypernetwork in which interaction descriptors modulate edge-specific message passing through FiLM-style conditioning, enabling direction-aware relational reasoning.
- Introduced multi-scale temporal encoding and deterministic/probabilistic forecasting variants, and evaluated the framework extensively on ETH/UCY and Stanford Drone Dataset through sparsity, efficiency, robustness, and physical-plausibility analyses. **Accepted at ACCV 2026 (Oral Presentation)**.

2025–2026 **Stochastic Activation Switching for Uncertainty Estimation**

- Co-developed Stochastic Activation Switching, a stochastic inference mechanism for uncertainty estimation that introduces controlled activation-level stochasticity as an alternative to conventional Monte Carlo Dropout.
- Investigated the calibration and predictive-uncertainty behavior of different stochastic activation formulations.
- Contributed to theoretical and empirical analysis of the method, including its sampling behavior, regularization effect, uncertainty-estimation performance, and ablation studies. **Accepted at BMVC 2026.**

2025–2026 **Precedence-Aware Mode Repair for Multimodal Trajectory Forecasting**

- Identified and formalized failure modes in multimodal trajectory forecasting where geometrically diverse predictions collapse to the same interaction precedence, introducing metrics for precedence coverage, order collapse, and bad-winner selection.
- Developed **PAMR**, a model-agnostic post-hoc repair framework for frozen multimodal forecasting models that generates precedence-targeted hypotheses through monotonic temporal warping and spatial trajectory correction.
- Evaluated PAMR across Argoverse 1/2, nuScenes, and Waymo Open Motion using multiple forecasting architectures, substantially reducing wrong-order hypothesis selection while preserving geometric accuracy. **Accepted at WACV 2027.**

2025–2026 **Intelligent Video Anomaly Detection**

- Developed hierarchical graph-based interaction modeling and anomaly-detection components for intelligent video surveillance, focusing on structured modeling of agent interactions and abnormal behavior.
- Contributed to the research design and technical development of the TÜBİTAK BİLGEM-funded project, integrating graph-based relational reasoning into the anomaly-detection pipeline.

2026–present **Topology-Aware Segmentation and Generative Refinement**

- Developed topology-aware learning and refinement methods for thin and curvilinear structure segmentation, including adaptive hypernetwork-based topology supervision and JEPA-style conditional generative refinement for repairing connectivity, branch, cycle, and fragmentation errors; evaluated using Dice, clDice, Betti-number error, and Betti matching.

2026–present **Pedestrian Intention Prediction**

- Investigating pedestrian intention prediction for autonomous-driving scenarios, with focus on modeling behavioral and contextual cues that indicate future crossing and motion intent before trajectory changes become directly observable.

Applied Deep Learning Lab, University of Tabriz *Oct 2023 – Sep 2025*
Research Assistant Tabriz, Iran

- Conducted research in pedestrian trajectory prediction and medical image analysis.

Grants, Funded Research & Intellectual Property

Intelligent Video Anomaly Detection System *2025 – 2026*
*TÜBİTAK BİLGEM-funded national research project (**€40,000**); Lead Researcher*

- Authored the technical proposal for the successfully funded €40,000 national research project.
- Led development of hierarchical graph-based interaction modeling and anomaly-detection components.

Invention Disclosure / Patent Protection *Ongoing*
Intellectual property arising from TÜBİTAK BİLGEM-funded research

- Invention disclosure under institutional evaluation for potential patent protection of research developed within the TÜBİTAK BİLGEM-funded project.

Teaching Experience

Bilkent University *Sep 2025 – Present*
Teaching Assistant, Department of Computer Engineering Ankara, Türkiye

- **Python Programming**
- **Digital Design**
- **Deep Generative Networks**

Supported laboratory sessions, assignments, grading, and student instruction.

University of Tabriz *Sep 2023 – Sep 2025*
Teaching Assistant, Department of Computer Engineering Tabriz, Iran

- **Introduction to Programming**
- **Data Structures and Algorithms**
- **Linear Algebra**
- **Data Mining**

Academic Service

- **Reviewer**, *Expert Systems with Applications*, Elsevier, 2025–2026.

Languages

English (IELTS 7.5); Persian (native); Azerbaijani (native).